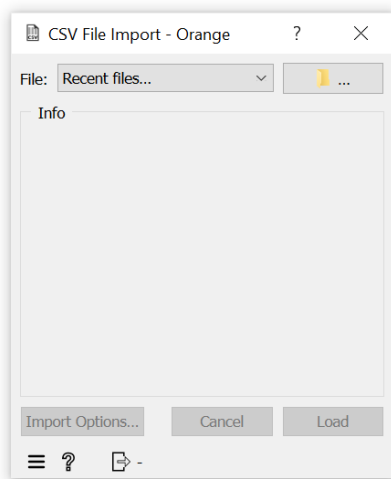
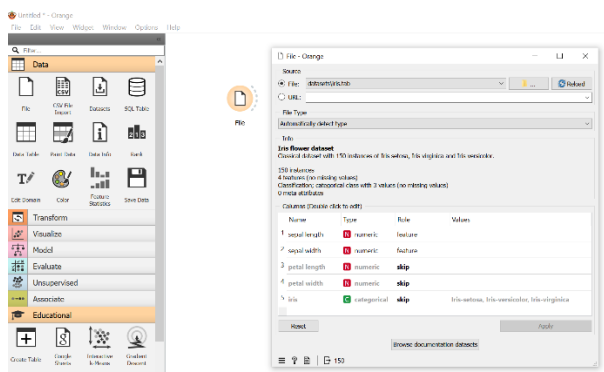
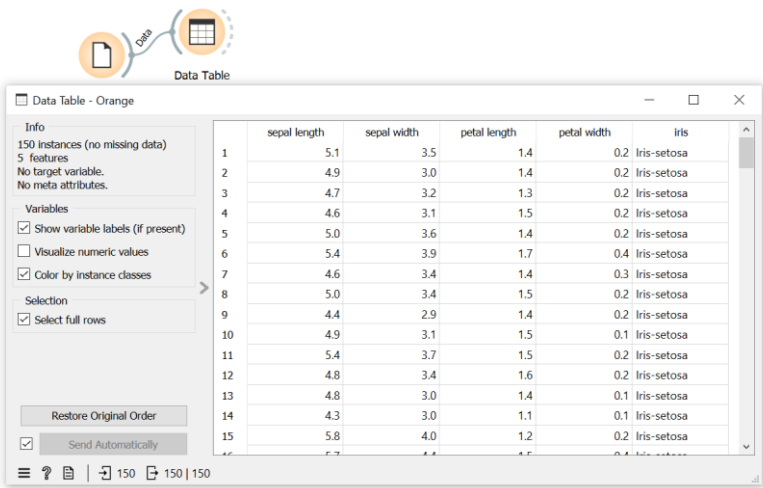


Manual simples do Orange:

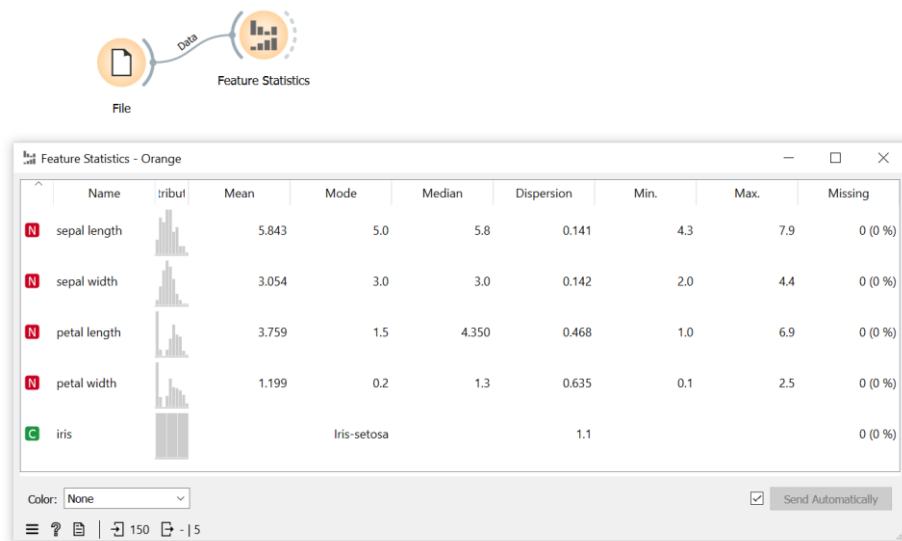
1. Obter dados:



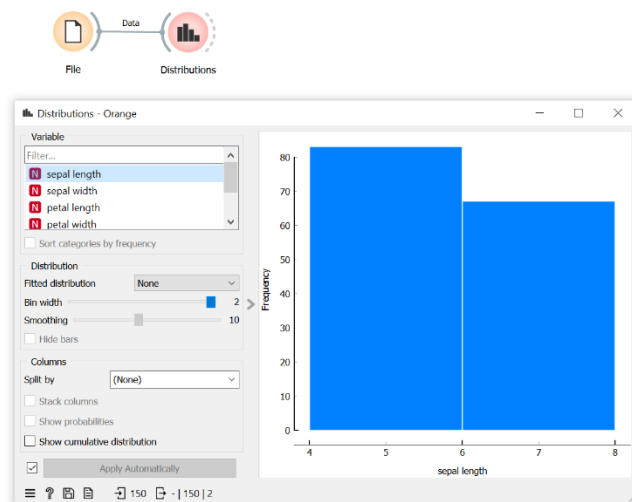
2. Tabela de dados:



3. Estadísticas de datos:

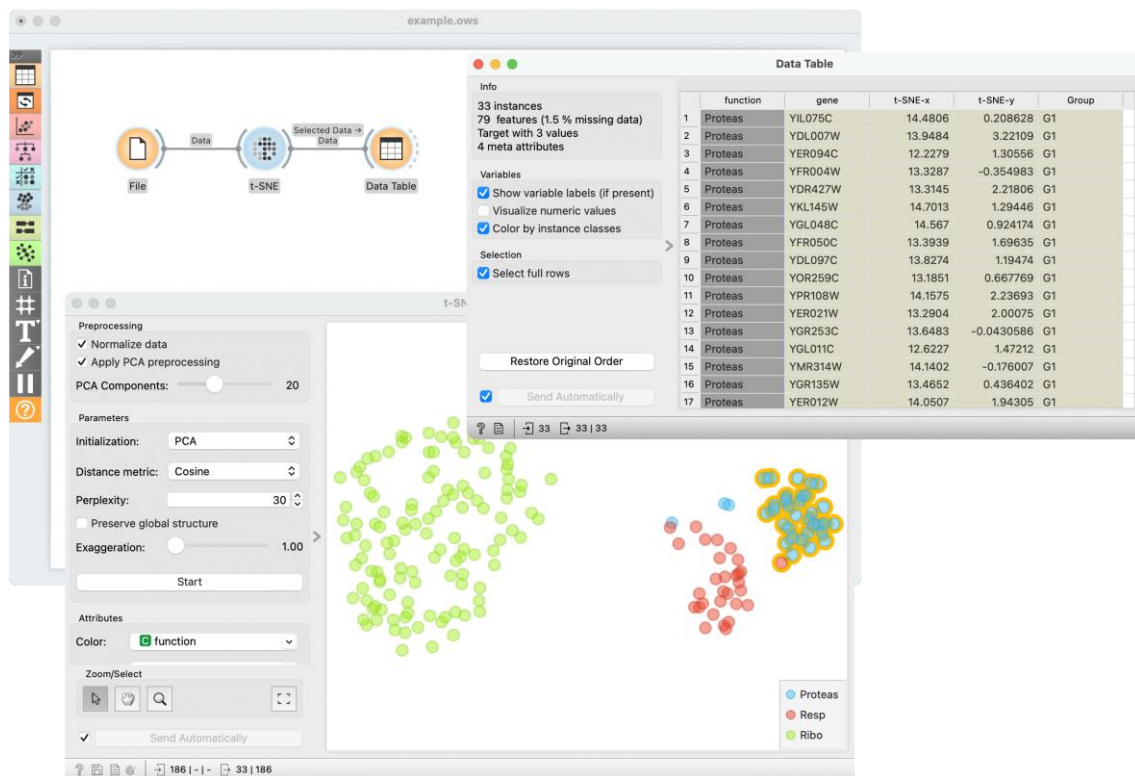


4. Gráficos de datos:

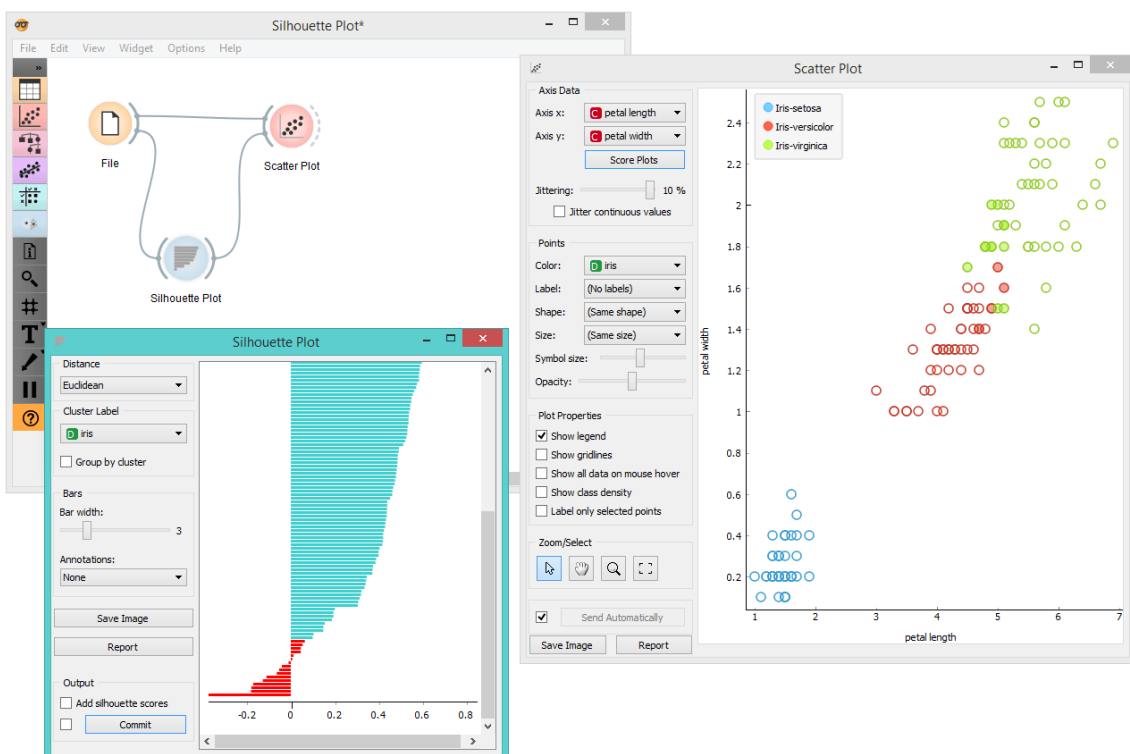


5. Redução de dimensão de dados:

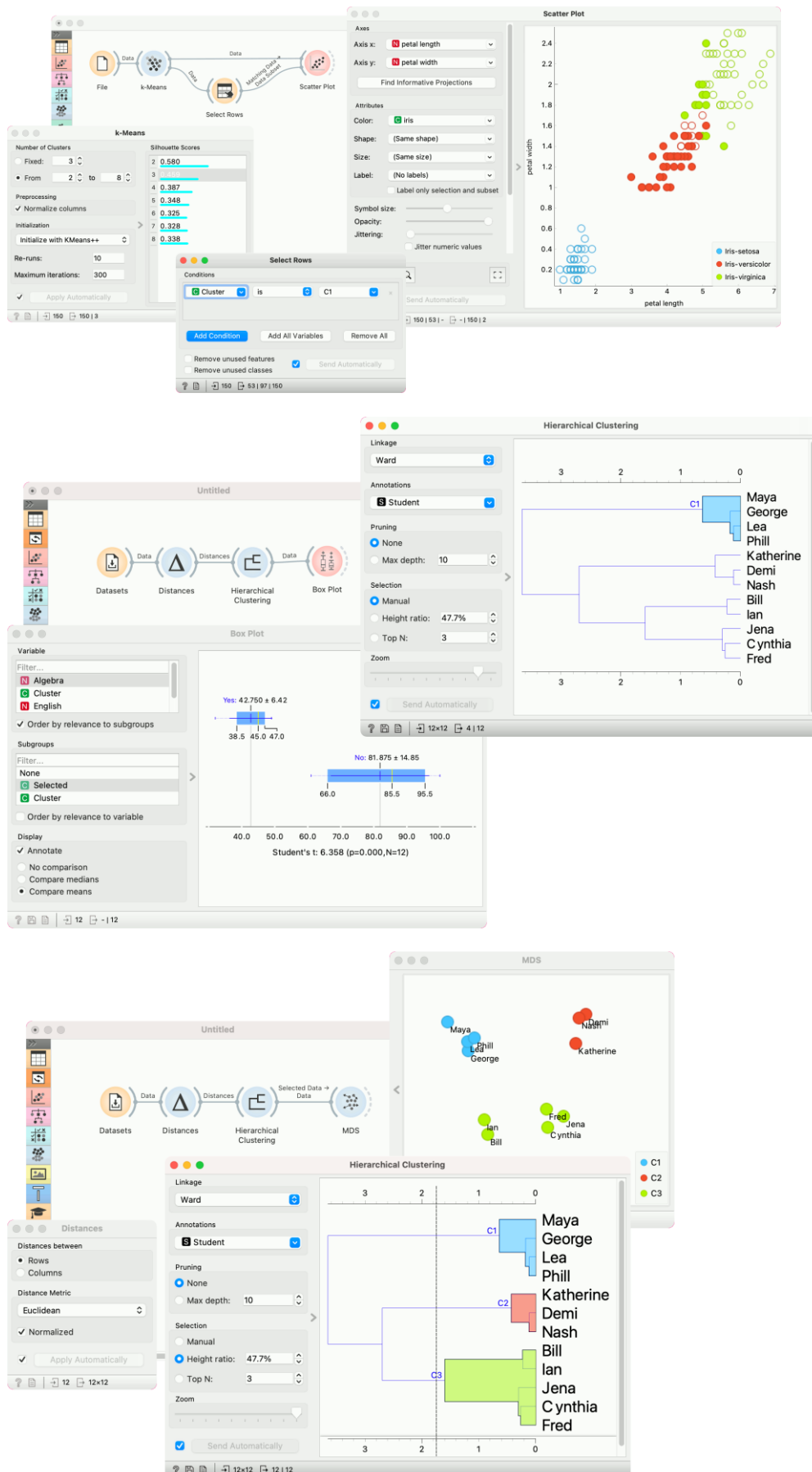




6. Determinar números de clusters:

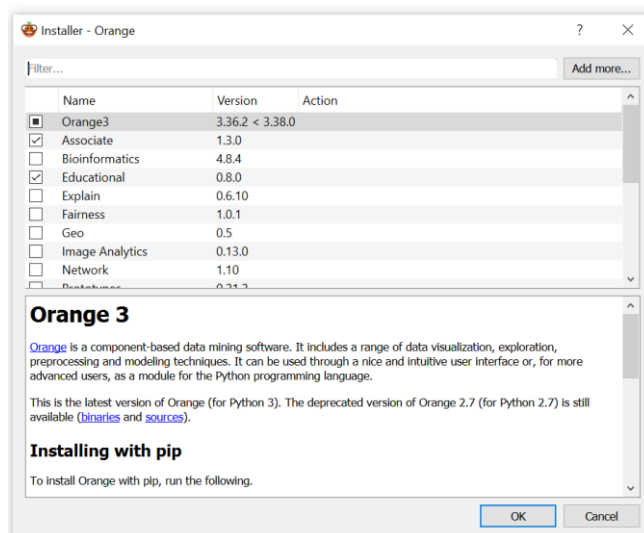
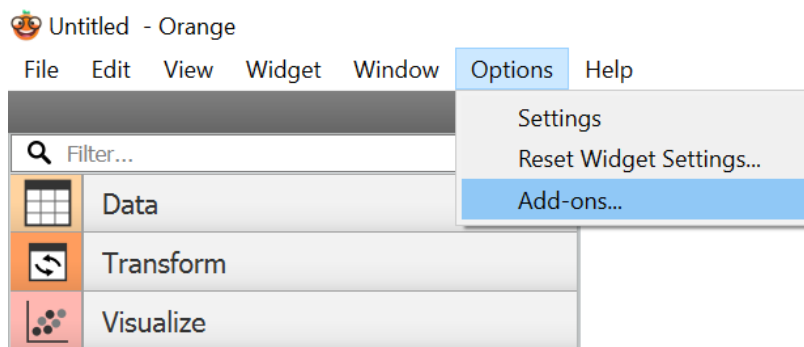


7. Agrupamento:





8. Regras de associação:



association rules*

File Edit View Widget Options Help

File Data Table Association Rules

Info

Number of rules: 10000
Filtered rules: 10000
Selected rules: 1
Selected examples: 56

Find association rules

Minimal support: 0.05%
Minimal confidence: 70%
Max. number of rules: 10000

☐ Induce classification (itemset → class) rules

☒ Auto find rules is on

Filter rules

Antecedent
Contains:

Min. items: 1 Max. items: 999

Consequent
Contains:

Min. items: 1 Max. items: 999

☒ Apply these filters in search

☒ Auto send selection is on

Supp	Conf	Covr	Strg	Lift	Levr	Antecedent	Consequent
0.01	0.70	0.01	31.41	2.74	0.00	tropical fruit, yogurt, root vegetables	whole milk
0.00	0.70	0.00	125.65	2.74	0.00	yogurt, other vegetables, bottled water, root vegetables	whole milk
0.00	0.75	0.00	209.42	2.94	0.00	yogurt, rolls/buns, bottled water, root vegetables	whole milk
0.00	0.75	0.00	314.12	2.94	0.00	yogurt, other vegetables, rolls/buns, bottled water, root vegetables	whole milk
0.00	0.77	0.00	146.38	3.98	0.00	yogurt, rolls/buns, soda, root vegetables	other vegetables
0.00	0.70	0.00	103.20	6.67	0.00	yogurt, whole milk, other vegetables, rolls/buns, bottled water	tropical fruit
0.00	0.88	0.00	171.50	5.27	0.00	tropical fruit, whole milk, other vegetables, rolls/buns, bottled water	yogurt
0.00	0.70	0.00	251.30	2.74	0.00	tropical fruit, yogurt, other vegetables, rolls/buns, bottled water	whole milk
0.00	0.70	0.00	190.30	3.62	0.00	tropical fruit, yogurt, whole milk, rolls/buns, soda	other vegetables
0.00	0.77	0.00	105.54	5.51	0.00	tropical fruit, rolls/buns, bottled water, soda	yogurt
0.00	0.83	0.00	228.67	5.97	0.00	tropical fruit, other vegetables, rolls/buns, bottled water, soda	yogurt
0.00	0.71	0.00	51.29	2.80	0.00	tropical fruit, yogurt, other vegetables, root vegetables	whole milk
0.00	0.81	0.00	93.07	3.19	0.00	tropical fruit, yogurt, rolls/buns, root vegetables	whole milk
0.00	0.72	0.00	139.61	2.83	0.00	tropical fruit, yogurt, other vegetables, rolls/buns, root vegetables	whole milk
0.00	0.70	0.00	51.60	6.67	0.00	yogurt, other vegetables, bottled water, root vegetables	tropical fruit
0.00	0.79	0.00	73.71	7.49	0.00	yogurt, whole milk, other vegetables, bottled water, root vegetables	tropical fruit
0.00	0.79	0.00	179.50	3.08	0.00	tropical fruit, yogurt, other vegetables, bottled water, root vegetables	whole milk
0.00	0.73	0.00	126.87	3.79	0.00	tropical fruit, yogurt, whole milk, bottled water, root vegetables	other vegetables
0.00	0.92	0.00	209.42	3.59	0.00	tropical fruit, rolls/buns, bottled water, root vegetables	whole milk
0.00	0.89	0.00	114.67	8.47	0.00	yogurt, whole milk, rolls/buns, bottled water, root vegetables	tropical fruit
0.00	0.73	0.00	124.73	5.21	0.00	tropical fruit, whole milk, rolls/buns, bottled water, root vegetables	yogurt
0.00	1.00	0.00	314.12	3.91	0.00	tropical fruit, yogurt, rolls/buns, bottled water, root vegetables	whole milk
0.00	0.83	0.00	228.67	5.97	0.00	tropical fruit, other vegetables, rolls/buns, bottled water, root vegetables	yogurt
0.00	0.83	0.00	418.83	3.26	0.00	tropical fruit, other vegetables, rolls/buns, bottled water, root vegetables	whole milk
0.00	0.83	0.00	172.00	7.94	0.00	yogurt, whole milk, other vegetables, rolls/buns, bottled water, root vegetables	tropical fruit
0.00	1.00	0.00	274.40	7.17	0.00	tropical fruit, whole milk, other vegetables, rolls/buns, bottled water, root vegetables	yogurt
0.00	1.00	0.00	502.60	3.91	0.00	tropical fruit, yogurt, other vegetables, rolls/buns, bottled water, root vegetables	whole milk
0.00	0.83	0.00	91.83	14.87	0.00	tropical fruit, other vegetables, rolls/buns, bottled water, root vegetables	yogurt, whole milk
0.00	0.71	0.00	153.14	6.55	0.00	tropical fruit, yogurt, whole milk, other vegetables, rolls/buns, bottled water	root vegetables
0.00	0.75	0.00	158.58	3.88	0.00	tropical fruit, rolls/buns, soda, root vegetables	other vegetables

frequent itemsets*

File Edit View Widget Options Help

File Data Table Frequent Itemsets

Info

9835 instances
169 features (tags, density 2.61%)
No target variable.
No meta attributes

Variables

☒ Show variable labels (if present)
☐ Visualize continuous values
☒ Color by instance classes

Selection

☐ Select full rows

Data Table

4	yogurt, pip fruit, cream cheese, meat spreads
5	whole milk, other vegetables, condensed milk, long life bakery product
6	yogurt, whole milk, butter, rice, abrasive cleaner
7	rolls/buns
8	other vegetables, rolls/buns, UHT-milk, bottled beer, liquor (appetizer)
9	pot plants
10	whole milk, cereals
11	tropical fruit, other vegetables, white bread, bottled water, chocolate
12	citrus fruit, tropical fruit, yogurt, whole milk, butter, bottled water, curd, flour, dis...
13	beef
	rolls/buns, frankfurter, soda
	tropical fruit, chicken
	butter, sugar, fruit/vegetable juice, newspapers
	fruit/vegetable juice
	packaged fruit/vegetables
	chocolate

Info

Number of itemsets: 122
Selected itemsets: 2
Selected examples: 3680

Expand all Collapse all

Find itemsets

Minimal support: 2%
Max. number of itemsets: 10000

☒ Auto find itemsets is on

Filter itemsets

Contains: soda

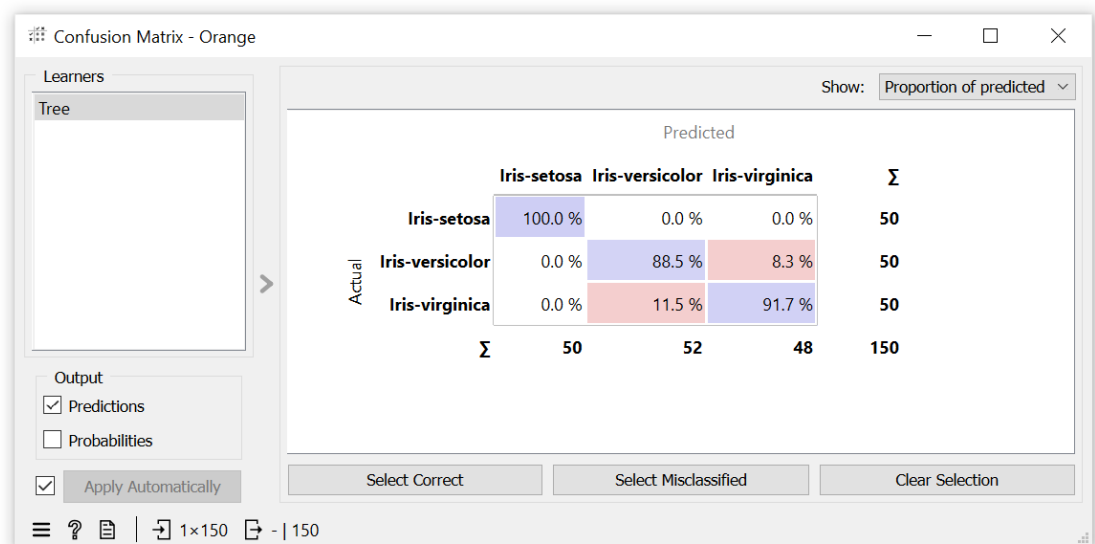
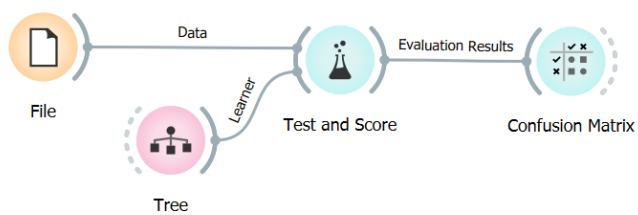
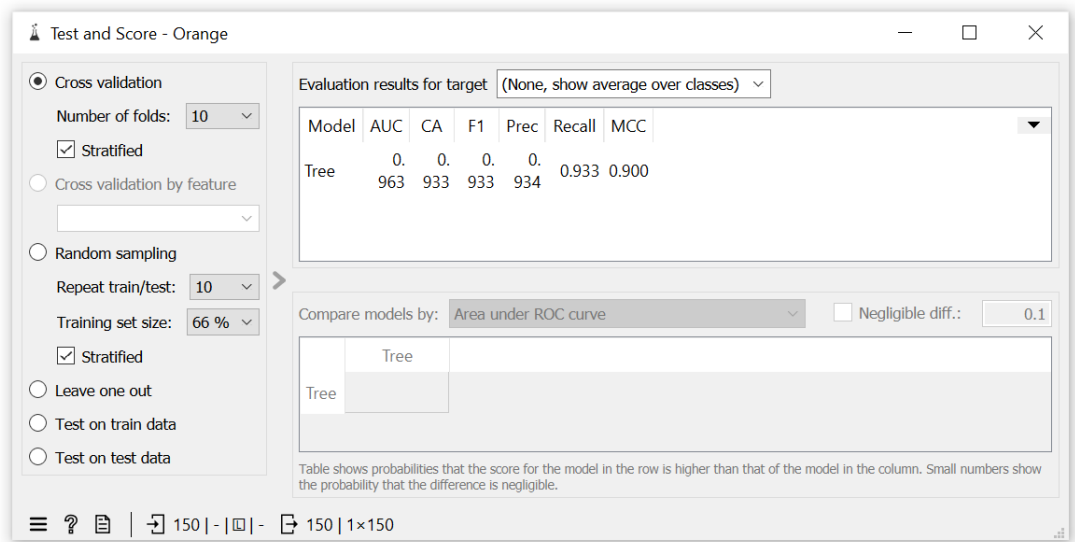
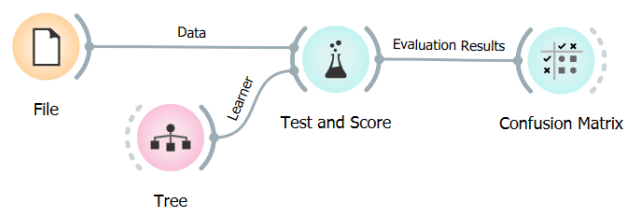
Min. items: 1 Max. items: 5

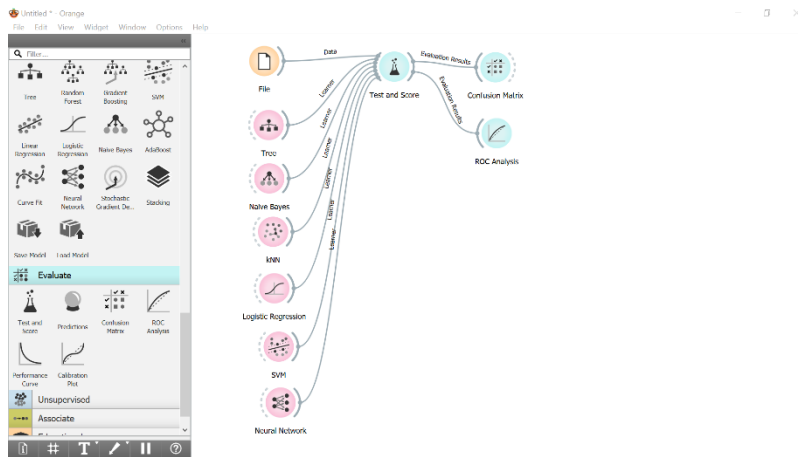
☒ Apply these filters in search

☐ Send selection

Itemsets	Support	%
whole milk	2513	25.55
soda	394	4.006
other vegetables	1903	19.35
soda	322	3.274
rolls/buns	1809	18.39
soda	377	3.833
soda	1715	17.44
shopping bags	242	2.461
sausage	239	2.43
pastry	207	2.105
yogurt	1372	13.95
soda	269	2.735
bottled water	1087	11.05
soda	285	2.898
tropical fruit	1032	10.49
soda	205	2.084
whole milk	2513	25.55
soda	394	4.006
other vegetables	1903	19.35
soda	322	3.274
rolls/buns	1809	18.39

Classificação:





Test and Score - Orange

☒ Cross validation

Number of folds: 10

☒ Stratified

☐ Cross validation by feature

☐ Random sampling

Repeat train/test: 10

Training set size: 66 %

☒ Stratified

☐ Leave one out

☐ Test on train data

☐ Test on test data

Evaluation results for target (None, show average over classes)

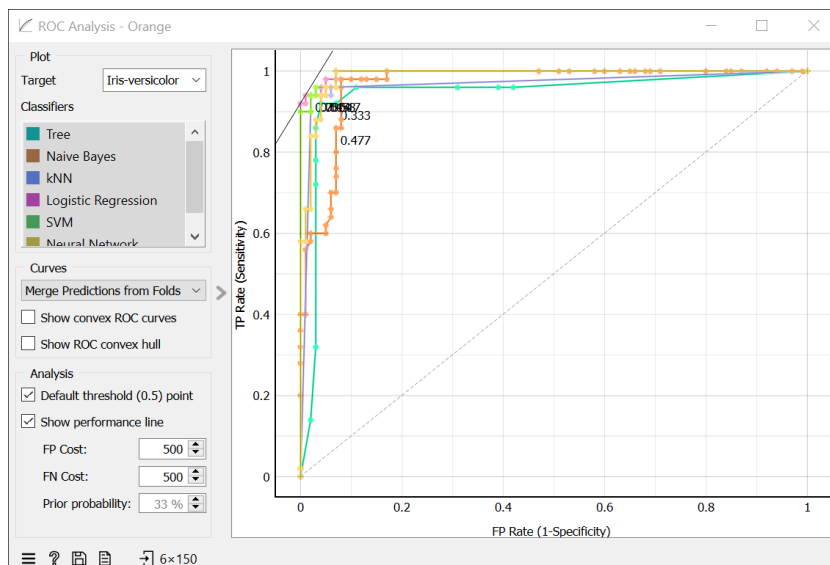
Model	AUC	CA	F1	Prec	Recall	MCC
Tree	0.963	0.933	0.933	0.934	0.933	0.900
Naive Bayes	0.981	0.893	0.893	0.894	0.893	0.840
kNN	0.979	0.960	0.960	0.960	0.960	0.940
Logistic Regression	0.998	0.967	0.967	0.967	0.967	0.950
SVM	0.997	0.960	0.960	0.960	0.960	0.940
Neural Network	0.993	0.947	0.947	0.948	0.947	0.920

Compare models by: Area under ROC curve

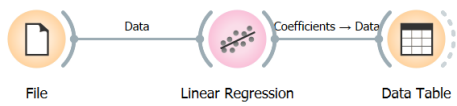
☐ Negligible diff.: 0.1

	Tree	Naive B...	kNN	Logistic ...	SVM	Neural ...
Tree		0.098	0.147	0.024	0.029	0.051
Naive Bayes	0.902		0.672	0.051	0.071	0.143
kNN	0.853	0.328		0.110	0.122	0.181
Logistic Regression	0.976	0.949	0.890		0.254	0.658
SVM	0.971	0.929	0.878	0.746		0.878
Neural Network	0.949	0.857	0.819	0.342	0.122	

Table shows probabilities that the score for the model in the row is higher than that of the model in the column. Small numbers show the probability that the difference is negligible.



Regressão:



Linear Regression - Orange

Name: Linear Regression

Parameters: ☐ Fit intercept (unchecking it fixes it to zero)

Regularization:
☒ No regularization Regularization strength:
☐ Ridge regression (L2) Alpha: 0.0001
☐ Lasso regression (L1) Elastic net mixing: L1 0.50 : 0.50 L2
☐ Elastic net regression

☒ Apply Automatically

Data Table - Orange

Info: 13 Instances (no missing data) 1 feature No target variable. 1 meta attribute

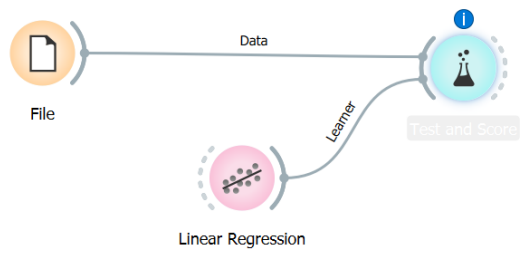
Variables: ☒ Show variable labels (if present) ☐ Visualize numeric values ☒ Color by instance classes

Selection: ☒ Select full rows

Restore Original Order

☒ Send Automatically

	name	coef
1	CRIM	-0.0928965
2	ZN	0.048715
3	INDUS	-0.00405998
4	CHAS	2.854
5	NOX	-2.86844
6	RM	5.92815
7	AGE	-0.00726933
8	DIS	-0.968514
9	RAD	0.171151
10	TAX	-0.00939622
11	PTRATIO	-0.392191
12	B	0.0149056
13	LSTAT	-0.416304



Test and Score - Orange

☒ Cross validation
Number of folds: 10
☒ Stratified

☐ Cross validation by feature

☐ Random sampling
Repeat train/test: 10
Training set size: 66 %
☒ Stratified

☐ Leave one out

☐ Test on train data

☐ Test on test data

Model	MSE	RMSE	MAE	MAPE	R2
Linear Regression	25....	5.056	3.381	0.178	0.697

Compare models by: Mean square error ☐ Negligible diff.: 0.1

	Linear Re...
Linear Regression	

Table shows probabilities that the score for the model in the row is higher than that of the model in the column. Small numbers show the probability that the difference is negligible.

Stratification is ignored for regression

Séries temporels:

